



S. No. ** 0098470 **

AIR UNIVERSITY, ISLAMABAD

ANSWER BOOK: MID SEMESTER EXAMINATION

REGISTRATION ID. 15982 CLASS & SEC 5-Semester SUBJECT Artificial Intelligence
S.NO. OF EXTRA SHEET/S (IF USED) 1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
EXAMINATION (Please ☒ the appropriate Box) ☒ FALL ☐ SPRING ☐ SUMMER 2025
SIGNATURE OF STUDENT [Signature] DATE 8-12-25
SIGNATURE OF INVIGILATOR [Signature] NAME OF INVIGILATOR Khalil Ur Rehman

IMPORTANT INSTRUCTIONS

BEFORE STARTING YOUR WORK, ENSURE THAT:

1. YOU ARE IN POSSESSION OF YOUR PRINTED ROLL NUMBER SLIP AND AU ID CARD.
2. YOU ARE SITTING IN THE ROOM AND SEAT ASSIGNED TO YOU AS PER THE ROLL NUMBER SLIP.
3. YOU DO NOT HAVE ANY UNAUTHORISED PRINTED OR WRITTEN MATERIAL, CELL PHONES, SMART DEVICE OR ANY OTHER MATERIAL THAT MAY BE AN UNFAIR AID AS PER AU ACADEMIC REGULATIONS. DISCOVERY OF ANY SUCH MATERIAL AFTER START OF THE EXAMINATION WILL RENDER YOU LIABLE FOR DISCIPLINARY ACTIONS UNDER CHAPTER XV OF AU REGULATIONS.
4. WRITE YOUR REGISTRATION ID AND PARTICULARS CAREFULLY ONLY IN THE SPACES SPECIFIED ABOVE FOR THIS PURPOSE.

DURING THE EXAMINATION:

5. ENSURE THAT ANSWER BOOK AND ALL CONTINUATION SHEETS ARE SIGNED BY THE INVIGILATOR.
6. ANSWERBOOK CONTAINS 10 PAGES. TEARING OF ANY PAGE FROM THE ANSWERBOOK OR CONTINUATION SHEET IS AN OFFENCE UNDER AU UNFAIR MEANS POLICY.
7. DO NOT WRITE ANYTHING ON QUESTION PAPER EXCEPT YOUR REG ID. USE LAST PAGES OF ANSWERBOOK FOR ROUGH WORK.
8. DO NOT BORROW ANYTHING FROM OTHERS. RAISE YOUR HAND TO SEEK ASSISTANCE OF THE INVIGILATOR.
9. DO NOT USE LEAD PENCIL (EXCEPT FOR DIAGRAMS/GRAPHS ETC.) OR ANY INK OTHER THAN BLUE OR BLACK.
10. WRITE ON BOTH SIDES OF EACH PAGE IN THE ANSWERBOOK.
11. ENSURE THAT SERIAL NUMBERS OF ANSWERBOOK AND CONTINUATION SHEETS ARE ENTERED ON ATTENDANCE SHEET. ALSO ENTER SERIAL NUMBER OF ALL CONTINUATION SHEETS IN THE SPACE PROVIDED ON THE ANSWERBOOK.

AFTER THE EXAMINATION:

12. MAKE SURE THAT ALL CONTINUATION SHEETS ARE STAPLED WITH MAIN ANSWERBOOK AT THREE PLACES BEFORE SUBMITTING YOUR ANSWER SCRIPT TO THE INVIGILATOR.
13. DO NOT LEAVE EXAM HALL WITHOUT PERMISSION OF INVIGILATOR EVEN IF YOU HAVE COMPLETED THE EXAM.
14. IN NO CASE SHOULD YOU LEAVE THE EXAM HALL BEFORE HALF TIME OR YOUR PAPER MAY BE CANCELLED.

Q. NO.	1	2	3	4	5	6	7	8	9	10	TOTAL
MAXIMUM MARKS	20	30									50
MARKS OBTAINED	16.5	21									37.5

MARKS OBTAINED IN WORDS Thirty Seven point Five

EXAMINER'S SIGNATURE [Signature]

Question # 01

(a) Artificial Intelligence:-

Artificial Intelligence (AI) means that computers and machines are act like a human. Artificial Intelligence (AI) systems can think, learn, decide, and act.

Types of AI :-

There are three types of AI given below :-

(1) Narrow AI :-

In this type of AI systems can't think and learn like humans.

(2) General AI :-

In this type of AI systems can think and learn like a human.

(3) Super AI :-

In this type of AI systems are more smarter than a human.

Purpose of Turing Test :-

Alan Turing is ~~is~~ also known as the Father of AI. He is born in 1912 and died in 1955.

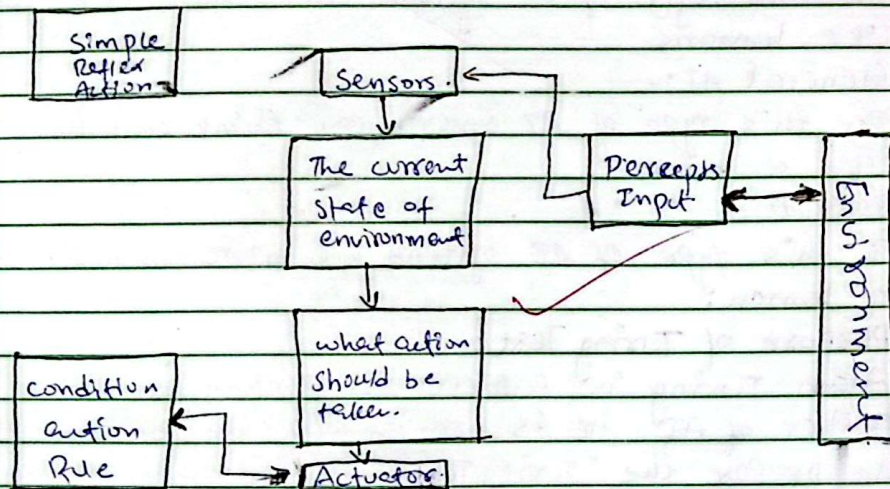
He proposed the "Turing Test" the purpose of Turing Test to make a computer and machine act like a human.

The Turing Test was performed

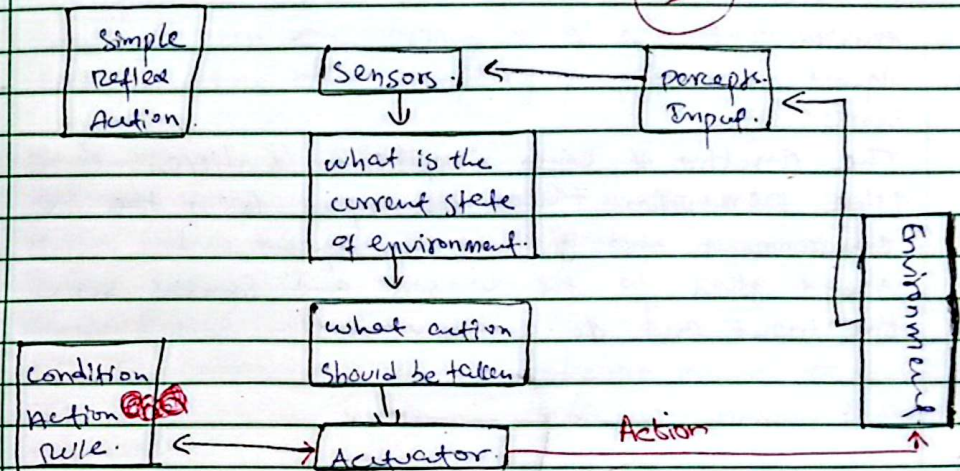
(a) Artificial Intelligence :-

Artificial Intelligence (AI) means that computers and machines are act like a human.

- (b) Agent in the context of AI:-
 An Agent is a anything that can perceive from its environment and act on it.
 Core Role:- Bridge between environment and Artificial Intelligence.
 Cycle:-
 perception $\rightarrow$ Decision $\rightarrow$ Agent $\rightarrow$ Environment.
 Diagram:-



- (c) Simple Reflex Agent and Goal-based Agent with a diagram:-
 Diagram:-



It is cleared by the diagram that an agent can perceive from its environment and given the input to sensors and sensors to Actuator and actuator to condition action rule. means that sensors sense the input and give the signal to Actuator and actuator give the signal to condition action rule that can perform a task or act on it.

Goal based agent Diagram?

(d) The Four Components of the PEAS model:-

the four components of PEAS model:-

Perception,

Environment

Agent

Sensor.

The function of each component is different
i.e. perception takes the input from the environment and gives to the Agent and Agent gives to the sensors and sensors sense the input and do a task or action on it.

Question # 02

(a) uninformed search:-

uninformed search is a searching algorithm that can work on the principle of Queue First in First out FIFO. The time complexity of uninformed search is $O(b)^d$.

Depth-First Search (DFS):-

Depth-First Search (DFS) is also a searching algorithm that can work on the principle of Stack Last in First out LIFO.

• Completeness:-

DFS is complete when the nodes are finite to search.

DFS is incomplete when the nodes are infinite to search.

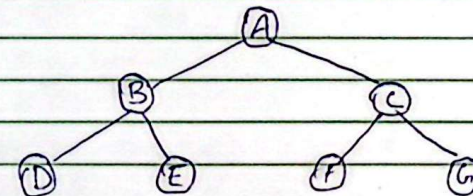
• optimality:-

DFS is optimal for positive step cost and DFS is nonoptimal for negative.

• Time Complexity:-

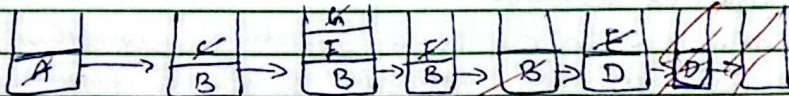
$O(b)^d$

Apply the DFS on the tree given below:-



Starting from the initial node "A" and the goal node is "G".

- DFS works on the principle of stack, LIFO so.
Apply the DFS on given tree-



The path is:-

A → B → C → F → G ✓

The time complexity is:-

$O(b)^d$

$$O(2)^2 \Rightarrow O(4) \quad \checkmark$$

(b) Uniform-cost Search (UCS):-

Uniform-cost Search (UCS) is also a searching algorithm that can work on the principle of priority Queue First-in-First-out (FIFO).

- Completeness:-

Uniform-cost Search (UCS) is complete-

- Optimality:-

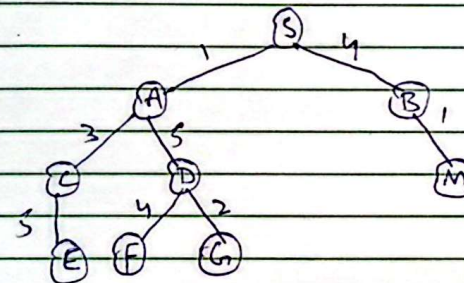
Uniform-cost Search (UCS) is optimal for positive step costs.

Uniform-cost Search (UCS) is non-optimal for negative step costs.

- Time complexity:-

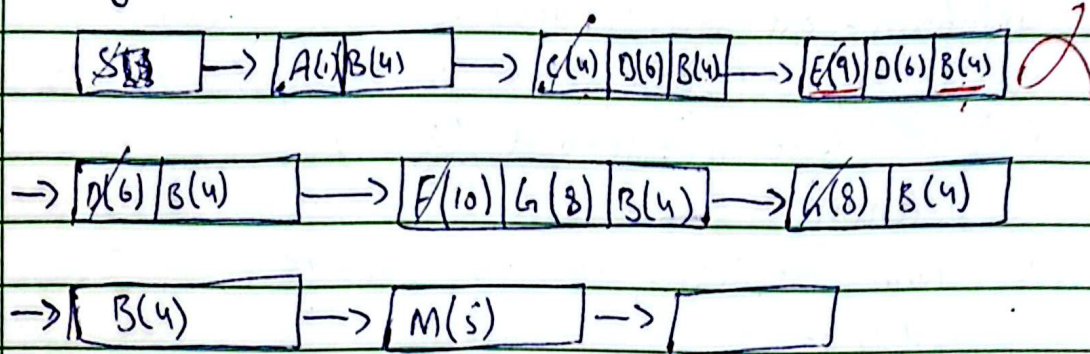
$$O(b^{nd})$$

Apply UCS on the tree given below:-



Starting from the Initial node 'S' and goal node is 'G'.

now apply the UCS on the given tree by using queue:-



The path by using queue node "S" to node "G" -

$S \rightarrow A(6) \rightarrow B(4) \rightarrow C(4) \rightarrow D(6) \rightarrow E(9) \rightarrow F(10) \rightarrow G(8)$

The Time complexity is:-

$$O(b^d)$$

$$O(2^3) \Rightarrow O(8)$$